

Nazanin Ghelichi

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SKILLS

Languages: Python, SQL, Excel **ML/DS:** scikit-learn, XGBoost, LightGBM, PyTorch, Hugging Face, SHAP
Data: pandas, NumPy, feature engineering, time-series **Tools:** Git, Streamlit, Google APIs, Power BI

EXPERIENCE

Founder & Math Tutor

2022 – 2026

MathoClock

- Built Study-Pal, an AI chatbot converting study intentions into tracked daily goals with automated reminders and leaderboard rankings; grew to 50+ active students.
- Designed streak and reward loops sustaining retention without manual intervention; owned full product lifecycle from user research to deployment.

Teaching Assistant

2022 – 2023

York University — Statistics, Calculus I & II, Linear Algebra I & II; ran weekly sessions resolving problems students couldn't crack in lecture.

Business Research Analyst

2020 – 2021

Diamond Bar and Co — Automated inventory tracking and analyzed customer behavior via Google Analytics to surface trends informing business decisions.

Student Researcher (NSERC USRA)

2019

York University

- Conducted original research on the Goldbach Conjecture under faculty supervision; awarded NSERC Undergraduate Student Research Award.

PROJECTS

IEEE Fraud Detection

github.com/nazaninghelichi/ieee-fraud-detection

- End-to-end fraud pipeline on 590k transactions; LightGBM + SHAP explainability, cost-matrix threshold tuning; ROC-AUC 0.964, PR-AUC 0.777, 88.5% recall.

MyNextBook — Hybrid Book Recommender

github.com/nazaninghelichi/MyNextBook

- Google Books API pipeline (1,300 books), TF-IDF + Sentence Transformer hybrid model, leave-one-out evaluation (Hit Rate@10: 0.290, NDCG@10: 0.141), deployed on Streamlit.

JobPal — Semantic Job Matching + RAG

github.com/nazaninghelichi/jobpal

- NLP pipeline for job-resume matching via embeddings and vector search; RAG layer for interview prep with hybrid retrieval and reranking.

Store Sales Forecasting

github.com/nazaninghelichi/store-sales-forecasting

- Retail time series forecasting with lag features, rolling means, temporal CV; demonstrates data leakage from random k-fold on time-ordered data.

Disaster Tweet Classification

github.com/nazaninghelichi/disaster-tweets

- Binary NLP classification comparing TF-IDF baselines against fine-tuned DistilBERT; error analysis on sarcasm and figurative language.

EDUCATION

M.Sc. Mathematics — York University

2023 – 2026

Thesis: *Extensive Analysis of Large Rank Elliptic Curves Over Function Fields* (Tate's algorithm, zeta-function methods, BSD conjecture)

B.Sc. Mathematics / B.Eng. Industrial Engineering (Dual Degree) — York University

2019 – 2022

Certifications: MIT Professional Education: ML & Data Science HarvardX: Business Analytics IBM: Excel Analytics